A 194-state Turing machine calibrated to the consistency of PA

Research and computational verification prepared for Vladimir Reshetnikov

13 November 2026, Pacific time

Abstract

We construct a deterministic, binary, one-tape Turing machine with **194 working states**, started on the all-zero tape, whose nonhalting is provably equivalent in PA to Con(PA). The state count excludes the separate terminal halt control. The construction keeps the supplied separate-diagonal-head numerical calculus and its entire 3,673-instruction register program. A new binary association of the unchanged 190-token axiom emitter admits 67 raw reader controls and seven compatible history-sensitive folds. Thus the complete count is $134 + 67 - 7 = 194$, improving both existing 195-state allocation patterns.

Every one of the 4,780 arbitrary-register source contracts passes, and literal blank execution reaches the exact complete source frame after 58,154 transitions. A separate symbolic literal-tape checker rederives its generic list lemmas from the actual table and independently checks every source branch, dense-table conjugacy and complete blank initialization. The archive contains every transition, deterministic builders, finite certificates, source and graph comparisons, search provenance, and this mathematical proof. A second 194-state table permutes the physical locations of registers A and D.

The proof establishes finite-trace equivalence with the supplied separate-head source and composes it with its explicit PA calibration: $\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_{194}) \leftrightarrow \mathrm{Con}(\mathrm{PA}))$. This is an ordinary mathematical proof of a PA-provability statement, supported by executable certificates. A newly compiled end-to-end Lean or Rocq theorem is not claimed. The comparison covers all 32 incoming archives at the pinned revision, recursively including nested archives, and distinguishes three positive 195-state families from six rejected diagnostic families.

Contents

1 The result and the literal machine **3**

2 Comparison of the supplied designs **3**

2.1 Pinned corpus and census method . . . 3

2.2 The three positive 195-state families . . . 4

2.3 What the larger designs contribute . . . 4

2.4 Rejected 195-state diagnostics . . . 5

3 The saving: 67 readers with seven compatible folds **5**

3.1 A fixed source program, a new binary association . . . 5

3.2 The seven folds in full . . . 6

3.3 Why this improves the two previous allocations . . . 7

4 Physical implementation and exact blank startup **7**

4.1 A 194-state register-allocation companion . . . 8

5 PA-internal finite-trace preservation **8**

6 Why the preserved source is calibrated to PA **9**

6.1 The numerical formulas and separate diagonal head . . . 9

6.2 Completeness: PA refutations produce source halts . . . 10

6.3 Soundness inside PA: source halts produce refutations . . . 11

7 Verification and the formal proof boundary **12**

8 Search provenance and failed approaches **13**

9 Relation to the ZFC and concise-iPRA ideas **14**

10 Reproduction and further research questions **14**

A Complete principal transition table **15**

B Archive-by-archive comparison **16**

C Source provenance **18**

1 The result and the literal machine

The principal machine is `construction/tables/pa194.tm`. Its working controls are $0, \ldots, 193$, its start is control 0, and its unique terminal halt label is 194. The tape is two-way infinite, has alphabet $\{0, 1\}$, and is initially zero everywhere, with the head at cell zero. Each transition writes one symbol and moves one cell left or right. There are **388 transition rows**; there is no row for the halt control. All 194 working controls are reachable in the pointed transition graph.

The native presentation starts in 74 and halts in 254. The supplied *dense-to-native* list is an exact conjugacy of every transition. The dense table's SHA-256 is

`4bfea62b17aad7d8c8c5e682fe8942e0a86b93e4351f8e675e32705bdaae007f`.

The native table's SHA-256 is

`e0a7397cf2134e8e7f58a8b1de2f7ff90b74ededd54c912f6ba5b7db48dabe54`.

The complete dense table appears in Appendix A. No alphabet change, nonblank input, stationary move or altered counting convention is used.

Design	Fixed controls	Raw AX readers	Folds	Total
Original PA195	134	68	7	195
Supplied 195S/195C	134	67	6	195
New M_{194}	**134**	**67**	**7**	**194**

The improvement is joint: keep the smaller 67-reader allocation and make seven folds simultaneously valid. Merely reducing the expanded source or counting individually promising folds does not give this result.

Let H_M be the usual arithmetic sentence asserting a finite halting trace of a literal machine from its specified blank start. The conclusion is

$$\boxed{\mathrm{PA} \vdash (\neg H_{M_{194}} \leftrightarrow \mathrm{Con(PA)}).}$$

Actual nonhalting follows if PA is consistent. The biconditional is a statement about what PA proves and does not add Con(PA) as an axiom. Graph reachability is not a claim that every control is visited by the blank execution. No global minimality claim is made for 194.

2 Comparison of the supplied designs

2.1 Pinned corpus and census method

The source corpus is the repository and incoming research at the pinned revision [1, 2]. The repository snapshot is

`5ed5a41fa5932433111eea096311364c6087e8e3`.

The inspected paths are the requested PA and ZFC research directories, `docs/incoming`, and the relevant concise-iPRA material. The incoming census recursively traverses ZIP members, deduplicating

archive payloads by SHA-256, and includes direct numeric `.tm` tables in the PA research directory. Complete numeric binary tables in archive `.tm` and `.txt` members are recognized when every working control has its two rows and exactly one terminal target is undefined.

The resulting counts are 32 top-level archives, 45 distinct archive payloads, 73 archive references, maximum nesting depth eight, 25,069 file occurrences, 2,360 recognized table occurrences, 582 byte-distinct tables, and 421 classified action-preserving graph classes. The counts include ancestors, experiments and rejected machines. They do not mean that there are 421 established consistency constructions. Two disconnected synthetic controls are excluded from the full-graph canonicalization; a separate no-halt synthetic control does not satisfy the table recognizer.

At 195 working states there are 37 occurrences, 14 byte forms and nine graph classes. Exactly three are positive manuscript constructions, while six are rejected diagnostics. No 194-state table was recognized. Tables embedded only in arbitrary source code or JSON are outside this numeric file census; the report and proof-layer review supplement the census. The full member-level inventory and all 32 archive contributions are in `comparison/`.

2.2 The three positive 195-state families

Family	Principal source choice	Readers−folds	Sites	Startup
Original PA195	Bundled diagonal-vacuity gate, original controller	$68-7$	3,700	58,154
195S	Separate diagonal head, original controller	$67-6$	3,673	58,154
195C	Separate diagonal head, changed Shift/MP controller	$67-6$	3,649	57,922

The original table is in `pa195_conpa_2026-11-13.zip`; 195S and 195C are in `PA195_ConPA_research_2026-11-14.zip`. The latter filename uses UTC dating and belongs to the pinned corpus even though this report uses Pacific dating.

The directories `pa195-quotient`, `pa195-all-tape`, `pa195-translate`, `pa195-ordinary`, `pa195-simulation`, `pa195-control`, `pa195-generic-trace`, and `pa195-global` are formal proof layers for the original 202-to-195 quotient. They are not eight additional machine designs. Their strengthened all-tape equivalence results are useful evidence about that quotient. Their scope does not turn the conditional PA endpoint into a completed machine-specific formal biconditional.

2.3 What the larger designs contribute

James Chang's PA209 and the U209 parent used by the U208 reports are different machines, despite their equal counts. The original U208 improvement is a history-sensitive control fold; later variants give 27 designated presentations in 20 rooted graph classes. They establish that several helper routings are possible, but their saving cannot be subtracted again from a descendant in which the same kind of sharing is already present.

PA196 changes the quantifier coding, axiom heads, axiom order and decoder allocation. Its original source has 3,812 primitive sites. The six-fold allocation is already reflected in 196. Later 196-state alternatives change axiom parenthesization, register placement, the Shift test, the MP result location, closing jumps and source paths. Several alter the same physical row and therefore cannot be combined by simply summing their advertised savings.

The diagonal-vacuity presentation reduces the emitter to 191 macro tokens. Separating the diagonal head reduces it to 190 tokens and changes the AX stack interface from nine pushed entries to ten. This logical simplification is the source of the principal machine. The 18-site direct-negation macro in the latest negative research packet is a useful counterexample to a naive metric: it reduces the expanded source to 3,441 sites but still needs 197 physical controls in its verified allocation.

The current comparison therefore has four distinct optimization levels: logical formulas and proof compilation; command-store operations; binary tree representation and compatible folds; and physical runtime and blank initialization. A claim at one level does not automatically establish an improvement at another.

2.4 Rejected 195-state diagnostics

Diagnostic family	Recorded obstruction
Two-rotation quotient	93 primitive-contract failures.
Warm physical candidate	714 failures; one failing site enters a four-step cycle.
Extended physical candidate	777 primitive-contract failures.
Seven-zero scanner regrouping	Wrong clean frame; an extra scanner control repairs the count to 196.
Short-scanner deletion	96 unrestricted failures and an incorrect invariant-conditioned return.
Zero-jump/helper fold	Wrong successor and wrong incremented register, also witnessed on the blank computation.

A failed proposed simulation rules out that certificate. It does not by itself prove that no unrelated interpretation of the table could ever have the desired property. Two rejected 196-state Shift designs have stronger negative evidence: both actually halt on blank tape after 43,563 steps. They are not consistency-machine candidates under the usual consistency assumption.

3 The saving: 67 readers with seven compatible folds

3.1 A fixed source program, a new binary association

The principal source is the supplied separate-head calculus used by 195S. It has an axiom-emission word of 190 macro tokens and a complete register program with 3,673 primitive instruction sites. The selected construction uses exactly that word in exactly that order. Its six independent axiom-head parameter permutations are all the identity; no axiom formula, guard, command selector, stack position or logical register assignment changes.

The frozen new grammar changes how that same word is parenthesized and shared as a binary tree. It has 67 raw reader controls. The 134 fixed controls are retained, giving a raw 201-state table, and the seven folds listed below reduce it to 194. The builder regenerates the parent 195S from its original grammar and folds and binds it to the exact supplied native table before checking the new construction.

Checking the flattened word alone would be insufficient. This tree language also encodes control flow: a `breakTo` or `continueTo` leaf counts ancestors of one direction, so reassociation can silently change its successor. The certificate therefore compares the entire resolved register instruction map, indexing sites by their left-to-right ordinal. It checks the opcode, register operand and every successor, including both decrement branches, every structured jump and every halt. The independent verifier uses a second interval/ancestor resolver for the same comparison.

Lemma 3.1 (Source identity)**.** *The new register program and the supplied separate-head program have identical executions from every initial vector of nine natural registers, when counters are identified by primitive-site ordinal. PA proves the corresponding finite-trace identity.*

Proof. Their complete finite instruction maps are equal. Hence their coded step functions make the same finite case distinction, register update and next-counter choice. Equality of the maps is a finite arithmetic fact; induction on the trace length gives the assertion in PA. □

A replayable occurrence-level reassociation certificate is included. It connects the parent and new macro trees through a common right comb. The certificate establishes endpoint word identity; intermediate trees are not asserted to meet the physical scanner constraints. The endpoint source-map and runtime certificates are checked separately.

3.2 The seven folds in full

The raw-to-folded identifications, in native compiler labels, are

$$2016 \mapsto 2052, \quad 2018 \mapsto 2027, \quad 2022 \mapsto 2032, \quad 2024 \mapsto 2046,$$
$$2037 \mapsto 2015, \quad 2055 \mapsto 2019, \quad 2060 \mapsto 2014.$$

Each pair has two distinct entry contexts. The removed member is entered after a symbol-preserving right move across zero; the retained member is entered after the corresponding move across one. The shared control can therefore recover the missing state information by inspecting the previous bit. The replacement actions are:

Removed	Retained	New read-0 action	New read-1 action
2016	2052	$0, R, 166$	$1, L, 2005$
2018	2027	$0, R, 90$	$1, L, 2020$
2022	2032	$0, L, 2002$	$1, R, 61$
2024	2046	$0, L, 2033$	$1, R, 2028$
2037	2015	$0, L, 2044$	$1, R, 2017$
2055	2019	$0, L, 2059$	$1, L, 2056$
2060	2014	$0, L, 2050$	$1, L, 2064$

All incoming targets are projected by the same seven identifications. Every other transition is retained from the raw 201-state construction. The pairs are disjoint and none contains the start or halt.

The local checker replays all four context/read-bit cases for each pair against the *final simultaneously folded table*. It requires the complete local tape, head displacement and projected destination control to agree, with arbitrary untouched exterior tape. This is 28 finite contracts. Helpers may be affected indirectly by other folds, so an optimistic matching of individually feasible pairs would be insufficient. The full source-frame checker then executes the resulting 194-state table for every primitive source case.

Proposition 3.2 (The new raw quotient)**.** *The raw 201-state table and its folded 194-state table have equivalent halting behavior from native state 74 on every initial tape. The local simulation uses between one and five folded steps for each raw step.*

Proof. All 23 incoming rows into exceptional fold controls establish the required previous bit. The start is not exceptional. At an ordinary control the transition commutes with the state projection;

at an exceptional control one of the 28 exact two-cell contracts applies. Thus induction constructs a positive finite simulation of each raw trace on its complete tape. For the converse, a folded first halt after T steps is inconsistent with $T+1$ nonhalting raw steps and their positive simulations, by determinism. The argument is also arithmetizable for coded initial tapes and traces. □

This quotient is of the newly compiled 201-state table. The result is not obtained by deleting one control from the literal supplied 195S table; transport from 195S uses the common source program.

The 28 local cases consist of ten one-transition cases, sixteen three-transition cases and two five-transition cases. The certificate also checks all 23 incoming rows into the exceptional controls. A constructive reassociation witness uses 369 occurrence-level rotations: 185 right rotations to the common comb and 184 inverse rotations to the new tree. Endpoint word equality and the complete resolved source graph are checked independently of these counts.

3.3 Why this improves the two previous allocations

The first positive PA195 kept 68 raw readers and saved seven controls by folding. The later 195S reduced the raw count to 67 but supported only six compatible folds. The new tree has the lower raw count and the larger fold set. It does not remove an already-counted fold a second time: its literal 201-state raw table and seven explicit identifications are reconstructed and counted from scratch.

An independent cross-combination first produced a useful fallback: deleting the outer diagonal-combination token from the original 195 grammar gave a separate-head realization with 68 readers and seven folds. The successful 194-state run, however, was seeded directly from the supplied 195S grammar, not from that crossed fallback. It focused on the best retained layouts with a lower annealing temperature and more frequent continuation from near-optimal candidates. This run found the 67-reader/seven-fold endpoint. The delivered frozen grammar, rather than the stochastic search, defines the final machine.

4 Physical implementation and exact blank startup

The source register order is $(N, T, W, A, D, H, X, Y, L)$. The principal machine keeps the supplied logical-to-physical map

$$(6, 2, 0, 4, 3, 7, 8, 5, 1).$$

A physical block $0\,1^{v+1}$ represents natural value v. The full frame consists of a left marker, the encoded binary source path, a gap, a right delimiter, the nine logical blocks in physical order and an arbitrary nonempty spare tail. For source path p the right delimiter is at distance $48+n$ from the source head, with $n \geq 0$ arbitrary; the zero gap therefore has length $48+n-|p|$. The verification uses this explicit frame family, not arbitrarily shorter headers. The arithmetic-box rows and fixed controller are unchanged. In particular, the inherited increment interface includes appending a one-valued spare register; omitting that effect would invalidate full-frame simulation.

Starting from the all-zero tape, literal execution reaches the exact clean frame after **58,154 transitions**, in native state 84, with head at -2, source counter 25 and logical vector

$$(2, 0, 0, 0, 0, 0, 0, 0, 0).$$

The endpoint comparison includes the complete finite support of the tape, not merely its head and control. It includes all markers, source bits and 26 one-valued spare blocks. An independent bytearray simulation verifies the same complete tape equality.

4.1 A 194-state register-allocation companion

The archive also supplies $M_{194\text{-}AD}$, using the same decoder and source graph but physical map

$$(6, 2, 0, 3, 4, 7, 8, 5, 1).$$

The A and D blocks are exchanged. In native labels the only three edits are

$$(17, 1) : 0R32 \mapsto 0R48, \quad (144, 1) : 0R48 \mapsto 0R32, \quad (161, 1) : 0R32 \mapsto 0R48.$$

The physical register arithmetic routines remain unchanged. This is a permutation of tape data locations, not a relabeling of controls. If σ exchanges file positions three and four, then $\sigma(v)_{p'(r)} = v_{p(r)}$ for every logical register r. Increment, zero test and positive decrement commute with this store identification, and the spare tail is unchanged.

The clean seed has $A = D = 0$ and keeps N at physical position six. The companion reaches its exact full clean frame after 58,122 literal steps. The 32-step saving is an initialization fact; no uniform speedup for all later computations is claimed. This register-layout technique was already used in a 196-state report [6]; the new contribution here is its verified composition with the 194-state decoder.

5 PA-internal finite-trace preservation

Let R_S be the common finite separate-head register program, started at site 25 with the clean logical vector above. Let H_R assert that it has a finite halting source trace. Both source and physical trace predicates have the usual primitive-recursive finite-trace checkers.

Lemma 5.1 (Exact positive simulation)**.** *Every nonhalting primitive step of R_S from a well-formed frame has a terminating, positive-length physical simulation by M_{194}. It ends at the exact intended next frame, applies the specified register update, preserves the remaining represented register values and untouched exterior tape, realizes the specified next-address and spare-tail effects, and has no premature halt. A source halt reaches the terminal physical control. These facts have PA proofs for the coded operations.*

Proof. All 4,780 source cases are checked, including both branches of every decrement. The symbolic frame includes an arbitrary natural parameter $n \geq 0$ in the header distance $48 + n$ and the arbitrary complete physical register file. The construction checker instantiates unchanged arithmetic-box contracts with their exact row premises; the independent checker rederives the generic register-list lemmas from the literal table and executes every source case.

The certificate language uses literal transitions, finite concatenations, constant-symbol scan loops and register-list scans and updates. Each has an elementary arithmetic correctness proof and a finite bound in the represented block lengths. The physical segment codes are primitive recursively constructible, with PA-provable correctness. The verifier's positive-progress condition requires a literal transition before a source contract can close; normalization steps do not count as physical progress. The final symbolic checker also provides a positive-duration certificate for its compressed list operations.

Replaying the finite certificates instantiates these arithmetic proofs. This argument quantifies over all frame parameters and does not infer a universal conclusion from bounded samples of register values. The source identity and independent successor checks supply the intended destination for every case. □

Theorem 5.2 (Finite-trace equivalence)**.** *For the principal table and the supplied separate-head machine M_{195S},*

$$\mathrm{PA} \vdash H_{M_{194}} \leftrightarrow H_R, \qquad \mathrm{PA} \vdash H_{M_{194}} \leftrightarrow H_{M_{195S}}.$$

Proof. Given a finite halting source trace, prepend the verified blank startup and concatenate its exact physical simulation segments. A recursion on the source trace constructs a finite halting tape trace; PA proves this construction correct by induction.

Conversely, suppose the physical run first halts after T transitions. The verified startup cannot contain a halt. Iterate the total source step function for $T + 1$ instructions from the clean seed, treating a halt as an absorbing event. If no source halt occurs, concatenate these positive-length nonhalting physical segments. Their combined length already exceeds T before adding startup. Determinism contradicts the alleged first physical halt. Thus the bounded source iteration contains a halt.

The bounded iteration and finite concatenations are primitive recursive; PA proves their correctness and the determinism contradiction by ordinary induction. The same argument for the supplied 195S eliminates the common source predicate. No assumption of actual nonhalting and no reflection principle for PA occurs in this preservation proof. □

6 Why the preserved source is calibrated to PA

This section makes the logical dependency explicit. The detailed supplied calibration is retained, with provenance, in `report/reference/`. We use its syntactic proof transformations, rather than inferring a PA theorem from external standard-model nonhalting.

6.1 The numerical formulas and separate diagonal head

For Cantor pairing

$$p(a, b) = \frac{(a + b)(a + b + 1)}{2} + b,$$

write

$$I(A, B) = 4p(A, B), \quad M(a, b) = 4p(a, b) + 1, \quad U(a, B) = 4p(B, a) + 2, \quad E(a, b) = 4p(a, b) + 3.$$

Code zero denotes truth, code one is the contradiction $M(0, 0)$, and $N(B) = I(B, 1)$ is negation. The quantifier arguments are transposed as shown. Conjunction and existential quantification have their usual classical implicational definitions.

The retained diagonal clause and the guarded old gate are

$$\begin{aligned}
\Gamma(B) &= B \to U(B, B),\\
V(a, b, c) &= M(a, b) \to U(c, M(a, b)),\\
P(a, b, c) &= c \to M(c, b),\\
S(a, b, c) &= U\big(b, (M(a, b) \to P(a, b, c)) \wedge (M(c, b) \to c)\big),\\
Q(a, b, c) &= U(c, S(a, b, c) \to N(P(a, b, c))),\\
G(a, b, c) &= N(V(a, b, c) \to Q(a, b, c)).
\end{aligned}$$

The separate-head AX command pushes $\Gamma(c)$ and leaves $G(a, b, c)$ as its top when $c \neq a, b$; the guarded top is truth otherwise. The diagonal head is safe without this guard: every variable label in formula

code B is strictly below B, so x_B is absent from its own formula. This follows by induction on the decoding, using strict component bounds for nonzero constructors and the separate truth-code case.

The other six universally instantiated heads are the PA196 heads:

$$\begin{aligned}
&(\exists x_a\, U(a, C)) \to C,\\
&U(a, U(b, C)) \to U(b, U(a, C)),\\
&M(c, a) \to (E(c, b) \to M(b, a)),\\
&E(c, a) \to (M(b, c) \to M(b, a)),\\
&\exists x_a E(c, a),\\
&U(a, b) \to \big(U(a, b \to c) \to U(a, c)\big).
\end{aligned}$$

There is also the unchanged dominated implicational head $((H \to d) \to D) \to ((D \to H) \to (A \to H))$ with the inherited numerical dominance condition $d \le D$, plus modus ponens and generalization of theorems. The finite command emitter is checked over a free algebra, including the order of the pushed formulas and the guard.

The bundled diagonal gate is

$$F_\Gamma(a, b, c) = N\big(\Gamma(c) \to (V(a, b, c) \to Q(a, b, c))\big).$$

Classical implicational templates convert F_Γ into both $\Gamma(c)$ and $G(a, b, c)$, and conversely. Every $\Gamma(B)$ is obtainable from a bundled gate with $c = B$ and $a = b = B + 1$. Thus the bundled and separate-head calculi have primitive-recursive proof translations preserving conclusions. The source-level separation changes the stack protocol, so the command compiler must account for ten pushed entries.

6.2 Completeness: PA refutations produce source halts

The supplied PA196 refutation compiler relies on two-way membership rewriting, the neighborhood-equality implication, and an equality witness. The diagonal calculus recovers these interfaces without assuming the removed equality-vacuity schema.

For a reverse rewrite, put

$$C = E(a, b) \to (\chi(b) \to \chi(a)), \qquad \varphi = N(C), \qquad c = \ulcorner \varphi \urcorner,$$

where $\chi(t)$ is a membership atom with t in either slot, or $M(t, t)$. First handle distinct labels, so the code bounds make the displayed separator label fresh. Pointed separation at (a, a, c) supplies $N(Q)$. The second half of its separation matrix and the propositional consequence $\varphi \to N(\chi(a))$ give

$$S \to U(a, M(c, a) \to N(\chi(a))).$$

The retained equality witness, direct membership rewriting and membership-only vacuity yield

$$S \to (M(c, b) \to N(\chi(b))).$$

Under φ, the point condition P supplies $M(c, a)$; direct rewriting supplies $M(c, b)$, whereas φ also supplies $\chi(b)$. Discharging assumptions gives $\varphi \to (S \to N(P))$. Generalization at c and the available diagonal clause $\Gamma(\varphi)$ yield $\varphi \to Q$, contradicting $N(Q)$. Double-negation elimination gives C. Coincident-label cases reduce to propositional reasoning and the already obtained diagonal membership case.

For neighborhood equality, first choose the fresh binder $z_0 = \ulcorner E(a,b) \urcorner$. The instance $\Gamma(E(a,b))$, universal distribution and the two-way membership rewrites give the neighborhood implication at z_0. The equality witness and membership-only vacuity transport the binder to each $z \neq a, b$. This also yields the neighborhood witness. These are the required replacement interfaces in the original compiler. Code construction, comparisons of labels and finite proof concatenation are primitive recursive, and PA checks their correctness.

For the ten-entry separate-head command, the adjusted stack words include

$$\begin{aligned} \text{pop} &= \mathsf{AX}(0,0,0)\,\mathsf{S}^{11}, \\ \text{push} &= \mathsf{AX}(0,0,0)\,\mathsf{S}^{9}\,\mathsf{MP}(0), \\ \text{discard} &= \mathsf{MP}(1)\,\mathsf{AX}(1,0,0)\,\mathsf{S}^{10}\,\mathsf{MP}(0). \end{aligned}$$

The generalization selector uses S^{10} after AX. The prior nine-entry selectors cannot be reused unchanged. The inherited exhaustive command-word scheduler, applied to this correctly adjusted compiler, therefore gives a PA-verifiable translation from each PA refutation into a finite halting source computation.

6.3 Soundness inside PA: source halts produce refutations

Interpret $M(a,b)$ by finite binary membership $\mathrm{Bit}(x_a, x_b)$, $E(a,b)$ by equality, and $U(a,B)$ by universal quantification of the translated formula. A convenient bit formula is

$$\mathrm{Bit}(s,z) \equiv \exists u, v\,[s = (2u+1)2^z + v \ \wedge \ v < 2^z].$$

PA proves $\mathrm{Bit}(s,z) \to z < s$, so it proves the negation of the translation of code one. The exceptional constructor equality $I(0,0) = 0$ is handled by a fixed proof of the equivalence of truth with truth implying truth; it is not treated as syntactic disjointness of constructors.

For every arithmetic formula $\theta(z, \vec{u})$, PA has a proof of finite binary comprehension

$$\forall n\, \exists s < 2^n\, \forall z < n\, (\mathrm{Bit}(s,z) \leftrightarrow \theta(z, \vec{u})).$$

The proof is induction on n. Retain the previous binary code if $\neg\theta(n, \vec{u})$, and add 2^n if $\theta(n, \vec{u})$. Substituting the syntax of θ into this fixed induction-proof pattern is a primitive-recursive proof-producing operation. PA verifies that it produces a proof. This statement concerns syntax and proof checking, not a uniform truth predicate for arithmetic.

For pointed separation with distinct labels a, b, the relevant finite comprehension matrix is

$$(\mathrm{Bit}(x_a, z) \vee z = x_b) \wedge T(c)[z/x_b],$$

with a suitable bound such as $x_a + x_b + 1$. When $a = b$, the binder changes both occurrences of that label; the correct matrix is instead $(z = x_b) \wedge T(c)[z/x_b]$. The first antecedent inside the binder is then $\mathrm{Bit}(z,z)$, which PA refutes. This distinction prevents a false free-variable treatment of the collision case. Code freshness provides the required separator label. The diagonal head and the other logical heads have uniformly generated first-order proof templates.

Annotate every visible theorem code, the cached head, and every live canonical stack entry with a PA proof of its translation. Both visible proofs matter because the original Shift halt test is $t + h = 1$. Canonical expansion of the Cantor stack treats $p(0,0) = 0$ explicitly: popping numerical zero gives the fixed truth proof; pushing a zero onto zero adds no live canonical cell. AX, MP and Shift preserve

this annotation by primitive-recursive proof transformations. A source halt produces a PA proof of the translated contradiction and hence a PA refutation. The scheduler carries its store, so this invariant is stated for every annotated store, without assuming a reset between command words.

This yields the two syntactic implications

$$\mathrm{PA} \vdash H_R \;\leftrightarrow\; \exists p\, \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner).$$

No soundness or reflection principle for PA is used inside PA.

Theorem 6.1 (Consistency calibration of the principal table)**.**

$$\boxed{\mathrm{PA} \vdash \left(\neg H_{M_{194}} \leftrightarrow \mathrm{Con}(\mathrm{PA})\right).}$$

Proof. Compose the last displayed source calibration with Theorem 5.2, then negate the existential PA-refutation predicate. The same finite-trace reasoning applies to the register-allocation companion, with its explicitly documented source-frame conjugacy. □

7 Verification and the formal proof boundary

Principal check	Result
Working controls / transition rows	194 / 388
Raw AX readers / simultaneous folds	67 / 7
Complete source instructions	3,673
Universal source branches, each implementation	4,780
Construction checker, total symbolic steps	1,479,348
Construction checker, maximum per branch	497
Independent checker, total symbolic steps	1,078,198
Independent checker, maximum per branch	397
Independent list lemmas rederived from literal rows	4
Minimum positive-duration constant	3
Exact complete-tape blank startup	58,154

All listed checks passed. The independent verifier uses no arithmetic-box oracle; its recorded branch certificate has SHA-256

`be3b1dd3ad86ac93dba48a40d04d8623ffc914cc3fbe84486a78548800e201c8`.

The companion passes the same 4,780 source cases in both implementations under its permuted physical map and reaches its full blank seed after 58,122 steps. Its independent checker takes 1,078,321 symbolic steps, with maximum 397 per outcome and minimum positive-duration constant three; it also rederives all four list lemmas from the literal transition rows.

The two physical checkers have independent representations and control-flow resolvers. The construction checker uses the documented arithmetic-box contracts after confirming their exact rows. The independent checker uses symbolic literal tape lists and rederives its generic list lemmas from the candidate table. Its parameters are nine independent arbitrary natural register values and an arbitrary finite nonempty list of spare values. Both decrement outcomes are covered. Their symbolic-step totals measure verification work, not physical runtime on arbitrary register inputs.

The release also checks all 388 rows, the 194 working controls, complete graph reachability, the unique halt target, exact dense-to-native conjugacy, the full source map, the 28 simultaneous local fold cases, whole-tape blank startup and the source-emitter word. The fresh corpus census contains no earlier recognized 194-state consistency table. The two new 194-state graphs are compared independently of control names.

The imported PA196 formal package exports the implication

$$\mathrm{PA} \vdash \mathrm{Nonhalt}(PA196) \to \mathrm{Con}(\mathrm{PA}).$$

It does not export a reverse PA-internal consistency implication for these new tables. The reverse direction used here comes from the explicit ordinary syntactic calibration summarized in Section 6 and retained in the reference files. The supplied original-PA195 formal quotient libraries prove increasingly strong all-tape preservation results, while its iPRA/PRA/PA endpoint assembly remains conditional on its machine-specific bridges.

Accordingly, the result delivered here is an ordinary mathematical proof of the stated PA biconditional, with executable machine-specific certificates and inherited logical proof constructions. The metadata marks the new endpoint as `kernel_checked:false`. A newly compiled end-to-end Lean or Rocq theorem is not being asserted.

8 Search provenance and failed approaches

The successful focused run started from the supplied 195S grammar and used random seed 2026111404. It first reported effective reader count 60 at proposal 4,454,453, after 229.826 seconds in the recorded run. The final candidate's score is 60 with 67 raw readers, seven compatible folds, zero scanner penalty and identity parameter permutations. The run was planned for 400 seconds and interrupted after the successful construction had passed verification. Its last complete cycle records 6,424,000 proposals at 329.167 seconds; no unrecorded final proposal total is asserted.

Two earlier 600-second runs completed 9,122,884 and 8,446,912 proposals from the supplied separate and original diagonal grammars, respectively, without improving effective count 61. A crossed-grammar run was interrupted with its last completed cycle recording 8,668,000 proposals at 544.531 seconds and best count 61. The exact source, numeric inputs, logs, winning grammar and run-status distinctions are retained in `search/`.

The successful search jointly scored raw decoder size and compatible folds. It used shared-subtree rotations, optional axiom-head parameter permutations, the physical scanner constraints and simultaneous local verification in its fitness. The selected 194-state result has identity parameter permutations, so its mathematical transport reduces to source identity and physical simulation. The search was stopped after preserving the successful construction and its certificates; no smaller-state claim is made.

Earlier searches in this session had reached distinct, fully verified 195-state alternatives. Those include a crossed 68-reader/seven-fold separate-head decoder and physical A/D register permutations. They were useful starting points and remain in the discovery record. The final 194-state theorem does not depend on the negative searches below.

The recorded failed searches include:

1. bounded-window control contractions and first-action synthesis, including alternative write bits, directions and existing controls;

2. two-, three- and four-bit reader-entry contexts, with a four-bit raw 67-reader search of 1,798 disjoint-context pairs recovering only the existing six pairs for that *particular* supplied tree;
3. contexts extracted from all arbitrary-register source contracts and complete startup of supplied 195S, yielding 8,758 radius-four windows over 172 non-box controls from 9,760 distinct symbolic frame observations;
4. complete closure of two fixed raw tables under a bounded local pair-contraction template, including alternate helper actions;
5. all $7! = 5,040$ orders of seven unconditional heads on each of two fixed tree shapes, with no sub-195 score for those shapes.

These are finite or heuristic searches in stated families, not lower bounds. The successful new tree itself demonstrates why failure on a fixed association cannot be promoted to a general obstruction.

Register permutations alone do not enlarge the current three-step pure-reader opportunity relation: their adjustable router rows write zero on reading one and move right, so they cannot be pure reader helpers or match either allowed AX reader-1 action. The reader-0 actions are fixed. This observation is limited to that fold language and says nothing against more general coupled changes.

The corpus also supplies concrete warnings against apparent reductions. An interior AX zero-run bound does not cover the first source path and transient increment frames, so it does not alone justify deleting a return scanner. Exposing membership vacuity as an unconditional stack head leaves unsound formulas available when the guard fails. Replacing the searched calculus by iPRA requires a new PA calibration, not just a smaller proof checker.

9 Relation to the ZFC and concise-iPRA ideas

The transferable ZFC principle is joint optimization of the source tree, reader allocation and physical helpers. A shorter source can eliminate a useful helper and increase the physical state count. The new 194 machine realizes the complementary outcome: one shared tree supports both the lower raw-reader count and the larger fold set. ZFC return/decoder substitutions also motivated the local action searches, although the accepted saving here occurs in the AX reader allocation.

The concise-iPRA material encourages reducing proof-checker interfaces by deriving some schemas and keeping the remaining dependencies explicit. The diagonal-code freshness simplification is already part of the supplied PA research; this report does not claim it as a new discovery or attribute it to a different theory without evidence. Proof-checker byte count and binary TM working-state count remain different objectives, and iPRA consistency is not interchangeable with PA consistency. The new machine keeps the existing PA proof calibration throughout.

10 Reproduction and further research questions

The top-level `README.md` gives exact commands. Python 3 and its standard library suffice to reconstruct the tables and replay the finite and symbolic certificates. Optional discovery searches require C++17. The report compiles with a standard LaTeX installation. The final frozen grammar and folds are included; stochastic discovery need not be repeated.

The most concrete next questions are:

1. Can a 66-reader layout preserve seven folds, or can a 67-reader layout support eight, giving 193 states with the fixed controller?
2. Can jointly changing a helper and several reader classes save a control beyond pairwise fold matching?
3. Can a three-context, two-history-bit decoder be certified as a whole replacement with a net state saving?
4. Can the complete outer control flow be reorganized to permit a shorter return scanner at every source and transient frame?
5. Can membership vacuity be split with a correctly guarded stack protocol whose physical cost is below the saved emitter cost?
6. Can the existing formal quotient and arithmetic-transport libraries be assembled into a kernel-checked endpoint for the exact 194-state table using these finite certificates?

A Complete principal transition table

Each entry is the write bit, movement direction and next control. H means the terminal label 194. The start is control zero on a blank-zero two-way-infinite tape. This table is generated directly from the delivered dense file.

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
0	1 R 1	1 R 2	65	0 R 91	1 R 49	130	0 R 134	1 R 135
1	1 R 3	1 R 1	66	0 R 92	0 L 75	131	0 R 136	1 R 137
2	0 R 0	1 R 2	67	0 R 86	1 R 33	132	0 R 138	1 R 139
3	0 R 4	0 R 1	68	0 R 93	1 R 68	133	0 R 140	1 R 141
4	1 R 5	1 L 6	69	0 L 94	1 L 33	134	0 R 100	1 R 67
5	1 L 7	1 L 8	70	0 L 84	0 R 2	135	0 R 142	1 R 127
6	0 L 7	1 L 6	71	0 R 95	1 R 71	136	0 R 143	1 R 144
7	0 L 9	1 L 6	72	0 R 75	1 R 96	137	0 R 145	1 R 146
8	0 L 10	0 L 11	73	0 R 59	0 R 85	138	0 R 134	1 R 147
9	1 R 12	0 L 9	74	0 R 86	1 R 97	139	0 R 120	1 R 148
10	0 L 9	0 L 10	75	0 R 42	0 L 13	140	0 R 149	1 L 131
11	1 L 8	1 L 11	76	0 R 87	1 R 37	141	0 R 150	1 R 151
12	0 R 12	0 R 13	77	0 R 30	1 R 98	142	0 R 79	1 R 41
13	0 L 14	0 R 13	78	0 R 42	1 R 99	143	0 R 28	1 R 152
14	1 L 15	0 L 15	79	0 R 100	1 R 101	144	0 L 153	1 L 154
15	0 L 15	1 L 16	80	0 R 102	1 L 103	145	0 L 129	1 R 155
16	0 L 17	1 L 16	81	0 R 30	1 R 80	146	0 R 156	1 R 157
17	0 L 18	1 L 16	82	0 R 97	1 R 104	147	0 R 149	1 R 158
18	0 L 19	1 L 16	83	0 R 30	1 R 105	148	0 R 159	1 R 120
19	0 L 20	1 L 16	84	1 R 84	1 L 15	149	0 R 100	1 R 160
20	0 L 21	1 L 16	85	0 R 106	1 R 85	150	0 R 161	1 R 162
21	0 L 22	1 L 16	86	0 R 73	1 R 107	151	0 R 163	0 L 10
22	0 L 23	1 L 16	87	0 L 46	0 R 93	152	0 R 164	1 R 165
23	0 R 23	1 R 24	88	0 R 108	1 R 98	153	0 R 166	1 R 157
24	0 R 25	1 R 26	89	0 R 108	1 R 69	154	0 R 167	1 R 156
25	0 R 27	1 R 28	90	0 R 109	0 L 14	155	0 R 135	1 R 130
26	0 R 29	1 R 30	91	0 R 110	1 L 74	156	0 R 147	1 R 41
27	0 R 31	1 R H	92	0 R 111	1 R 112	157	0 R 134	1 R 168
28	0 L 32	1 R 33	93	0 R 113	1 R 93	158	0 R 41	1 L 169
29	0 R 34	1 R 35	94	0 R 50	1 R 103	159	0 L 170	1 R 171
30	0 L 36	1 L 37	95	0 R 114	1 R 95	160	0 R 91	1 R 64

q	Read 0	Read 1	q	Read 0	Read 1	q	Read 0	Read 1
31	0 R 38	1 R 39	96	0 R 30	1 R 49	161	0 R 172	1 R 170
32	0 R 40	1 R 41	97	0 R 48	1 R 61	162	0 R 173	1 R 174
33	0 R 42	1 R 43	98	0 R 87	1 L 115	163	0 R 175	1 R 176
34	0 R 44	1 R 45	99	0 R 104	0 L 14	164	0 R 177	1 R 178
35	0 R 46	1 R 47	100	0 R 36	1 R 116	165	0 L 179	1 L 180
36	0 R 48	1 R 49	101	0 R 115	1 R 78	166	0 R 149	1 R 140
37	0 R 50	1 R 51	102	0 L 46	0 R 113	167	0 R 165	1 R 181
38	0 R 52	1 R 42	103	0 L 45	1 R 117	168	0 L 130	1 R 120
39	0 R 53	1 R 54	104	0 L 94	1 R 118	169	0 R 182	1 R 135
40	0 R 55	1 R 56	105	0 R 119	0 L 14	170	0 R 183	1 R 145
41	0 R 57	1 R 58	106	0 R 71	1 R 106	171	0 R 158	1 R 130
42	0 R 59	0 R 60	107	0 R 120	1 R 30	172	0 R 139	1 R 165
43	0 R 61	0 L 14	108	0 L 46	0 R 106	173	0 R 158	1 R 184
44	0 R 62	1 R 63	109	0 R 70	0 R 95	174	0 R 185	1 R 91
45	0 R 55	1 R 64	110	0 R 59	0 R 95	175	0 R 31	1 R 53
46	0 L 65	1 R 66	111	0 L 65	1 R 121	176	0 R 186	1 R 175
47	0 R 36	1 R 67	112	0 R 122	1 R 123	177	0 R 134	1 R 142
48	0 R 59	0 R 68	113	0 R 14	1 R 113	178	0 R 32	1 R 187
49	0 R 69	0 L 14	114	0 R 68	1 R 114	179	0 R 144	1 R 188
50	0 R 70	0 R 71	115	0 R 124	1 L 67	180	0 R 189	1 R 190
51	0 R 72	0 L 49	116	0 R 125	1 R 126	181	0 R 41	1 R 191
52	0 R 73	1 R 54	117	0 R 120	0 L 14	182	0 R 168	1 R 41
53	0 L 74	0 L 15	118	0 R 94	1 L 98	183	0 R 192	1 R 120
54	0 R 75	0 L 10	119	0 R 127	1 R 127	184	0 R 153	1 R 146
55	0 R 76	1 R 77	120	0 R 70	0 R 68	185	0 R 165	1 R 180
56	0 R 78	1 R 79	121	0 R 128	1 R 129	186	0 R 96	1 R 115
57	0 R 80	1 R 81	122	0 R 32	1 R 36	187	0 R 193	1 R 159
58	0 R 82	1 R 83	123	0 R 31	1 R 79	188	0 R 169	1 R 154
59	0 L 84	0 R 1	124	0 R 59	0 R 114	189	0 R 158	1 R 192
60	0 R 85	1 R 60	125	0 R 102	1 R 117	190	0 R 179	1 R 182
61	0 L 76	1 L 86	126	0 R 82	1 R 81	191	0 R 182	1 R 41
62	0 R 87	0 L 7	127	0 R 120	1 R 120	192	0 R 149	1 R 168
63	0 R 88	1 R 89	128	0 R 130	1 R 131	193	0 R 79	1 R 144
64	0 R 42	1 R 90	129	0 R 132	1 R 133			

B Archive-by-archive comparison

Archive	Main contribution and status
`Below_U208_Research_2026-10-29.zip`	Full-frame windows, offset quotients and guarded pruning; source 3,909→3,904; no accepted below-208 table.
`ConPA_below208_research_2026-10-29.zip`	Finite-model role-collapse objection, 279,936 fixed-association parameter cases, fold synthesis; no sub-208.
`PA_consistency_TM_research_2026-11-13.zip`	159-node five-zero source and 157-node combined representation; exact source CFG checks, no smaller physical allocation.
`U208_ConPA_2026-10-29.zip`	Particular native 172/211 history fold reduces a U209 parent to U208.
`U208_ConPA_Conditional_Context_2026-10-29.zip`	Conditional-entry proof; same two original U208 graph classes.
`U208_ConPA_Entry_Context_2026-10-29.zip`	Full entry-context proof; same two original classes.
`U208_ConPA_History_2026-10-29.zip`	Finite-history proof with four exceptional blocks; same original saving.

Archive	Main contribution and status
conpa_tm_comparison_and_reduction_search.zip	Experimental MP-result-in-H and MP-without-zero-push 196 variants; rejected Shift layouts halt at 43,563.
conpa_tm_comparison_and_search.zip	A/D register-swap 196; rejected two-rotation 195 fails 93 contracts.
conpa_tm_study_2026-11-13.zip	Combined 196 = H-only Shift + MP-result-in-H: 3,788 sites, 4,928 contracts, 57,922 startup; direct-negation 197 too.
pa196_research_2026-11-13.zip	Exact 157-node representation of 3,812-site original program; loses contextual sharing, no physical reduction.
pa208_research_2026-10-29.zip	Eighteen further helper layouts; 27 designated U208 presentations give 20 graphs, not two.
pa_below196_research.zip	Eight equivalent gates, membership/equality orientations, broad grammar searches; original 201-token best 196.
pa_consistency_comparison_2026_11_13.zip	Modal-head/explosion split, twelve semantic compositions, joint gate; physical results 196–200.
pa_consistency_comparison_and_new196_2026-11-13.zip	Diagonal 196 (3,700 sites), Separate 196 (3,673), Scanner-best 196; principal logical/source parents of positive 195.
pa_consistency_reduction.zip	Rebracketed 196, H-only 196, integrated gate 203/206; two failed 195 diagnostics.
pa_consistency_research.zip	Two exact 158-node sources and repaired 196; nominal 195 needs added scanner state.
pa_consistency_research_2026-10-29.zip	Inductive 3,909→3,904 specialization; no physical sub-208.
pa_consistency_research_2026-11-13(2).zip	Twelve-rotation 196, diagonal 198, successful-MP halting-source experiment; no sub-196.
pa_consistency_research_2026-11-13.zip	Earlier diagonal-vacuity 199 and short-negation 197.
pa_tm_comparison_and_reductions.zip	Two rotated 196 graphs; distributed-gate 198 with 3,773 source sites.
pa_tm_extended_research_2026-11-13.zip	Four closing-decrement variants and stronger zero invariants; all change the same row, hence not stackable.
pa_tm_research_2026-11-13(2).zip	Zerojump 196 (4,959 contracts, 58,402 startup); parity/disequality experiments; rejected 195 fails source 26.
pa_tm_research_2026-11-13.zip	Shorter-path 196 and independent physical checking; detailed direct PA reverse-translation strategy.
pa_unary_power_cover.zip	Unary weak-covering theory and PA proof translations; no new compiled small table.
u208_comparison_research_2026-10-29.zip	Original two-class audit; weak/exact schemas; rejected 207.
u208_research_2026-10-29.zip	Primary/alternative U208 and symbolic contracts; same initial history fold.
pa195_conpa_2026-11-13.zip	Positive original PA195: unchanged diagonal 191-token program, 68−7 reader allocation, detailed ordinary PA calibration.
PA195_ConPA_research_2026-11-14.zip	Positive 195S and 195C: unchanged separate 190-token AX word, 67−6 readers; full source/runtime/startup certificates plus independent literal-tape audits.
pa_consistency_cross_archive.zip	Four new 196-state combinations: diagonal/separate AX + Combined controller, each optionally zero-jump; sites 3,676/3,649 and startups 57,922/58,168.
pa_consistency_reduction_audit_2026-11-14.zip	Four further experimental MP-H compositions, two 196-state and two 197-state; parameter/order/association/helper/scanner negative searches.
pa_consistency_research_2026-11-14.zip	NEG18: `INC7; INC7; node64`, 18-site encoded negation; complete source 3,441 and 197 states (134+68−5); independent macro and scratch-zero proofs.

C Source provenance

References

[1] Vladimir Reshetnikov, *ProveIt*, pinned revision `5ed5a41fa5932433111eea096311364c6087e8e3`. https://github.com/VladimirReshetnikov/ProveIt/tree/5ed5a41fa5932433111eea096311364c6087e8e3/Computability/BusyBeaver/Research/PA.

[2] ProveIt, incoming research corpus at the same revision. https://github.com/VladimirReshetnikov/ProveIt/tree/5ed5a41fa5932433111eea096311364c6087e8e3/docs/incoming. The exact archive and member hashes are in `comparison/census.json`.

[3] James Chang, *Nonhalting and PA-independence for a 196-state Turing machine*, imported upstream revision `407cdc356f6b586635e71ac92be394255f4bfea4`. https://github.com/jamesyc/BusyBeaver/tree/407cdc356f6b586635e71ac92be394255f4bfea4.

[4] `pa195_conpa_2026-11-13.zip`, original 68-reader/seven-fold PA195 and its mathematical PA calibration, in the pinned incoming corpus.

[5] `PA195_ConPA_research_2026-11-14.zip`, separate-head 195S/195C constructions, portable compiler, independent physical verifier and complete older-archive comparison, in the pinned corpus.

[6] `conpa_tm_comparison_and_search.zip`, 196-state A/D register-allocation variant, in the pinned incoming corpus. The current construction transfers this physical layout idea to the verified 194-state decoder.

[7] ProveIt, ZFC consistency-machine research, at the pinned revision. https://github.com/VladimirReshetnikov/ProveIt/tree/5ed5a41fa5932433111eea096311364c6087e8e3/Computability/BusyBeaver/Research/ZFC.

[8] ProveIt, concise iPRA research, at the pinned revision. https://github.com/VladimirReshetnikov/ProveIt/tree/5ed5a41fa5932433111eea096311364c6087e8e3/Logic/PrimitiveRecursiveArithmetic/ConciseIPRA.